

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B4: SUB-ATOMIC PHYSICS

TRINITY TERM 2019

Monday, 10 June, 2.30 pm – 4.30 pm

*Answer **two** questions.*

*Start the answer to each question in a **fresh book**.*

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

1. (a) (i) Explain what is meant by a *Nuclear Form Factor*, $F(q)$. [2]
(ii) Show that for a spherically symmetric charge distribution

$$F(q) = -\frac{4\pi}{q} \int r\rho(r) \sin(qr) dr,$$

explaining the meaning of the symbols $\rho(r)$ and q . Modelling the nucleus as a uniformly charged ball of radius R , show that $F(q)$ may be expressed in the form

$$F(q) = C \left(\frac{1}{x^2} \cos x - \frac{1}{x^3} \sin x \right),$$

and give expressions for C and x . [10]

(b) An experiment is performed in which 250 MeV electrons are scattered from two different targets, one made of $^{40}_{20}\text{Ca}$, and the other of $^{48}_{20}\text{Ca}$. For each target the angle relative to the beam at which the scattered electrons are observed is gradually increased and the position of the first pronounced minimum in the scattered flux is recorded.

(i) For one of the targets the first minimum occurs at an angle of 53° whereas for the other the corresponding minimum is at 57° . Use the scattering measurements to determine the radius of each of the two Ca nuclei. [10]

(ii) To what extent do these scattering measurements support the liquid drop model of the nucleus? [3]

2. (a) (i) Explain how the spin, parity and strangeness of the lightest pseudoscalar octet of mesons may be explained within the quark model. [6]

(ii) The table below shows some measured branching fractions for the neutral pion.

Mode	Branching ratio
$\pi^0 \rightarrow 2\gamma$	98.8%
$\pi^0 \rightarrow e^+e^-\gamma$	1.17%
$\pi^0 \rightarrow e^+e^-e^+e^-$	3.3×10^{-5}
$\pi^0 \rightarrow \nu\bar{\nu}$	$< 3 \times 10^{-7}$

Draw one leading-order Feynman diagram for each decay, and hence explain the order of magnitude of each of these branching ratios. [8]

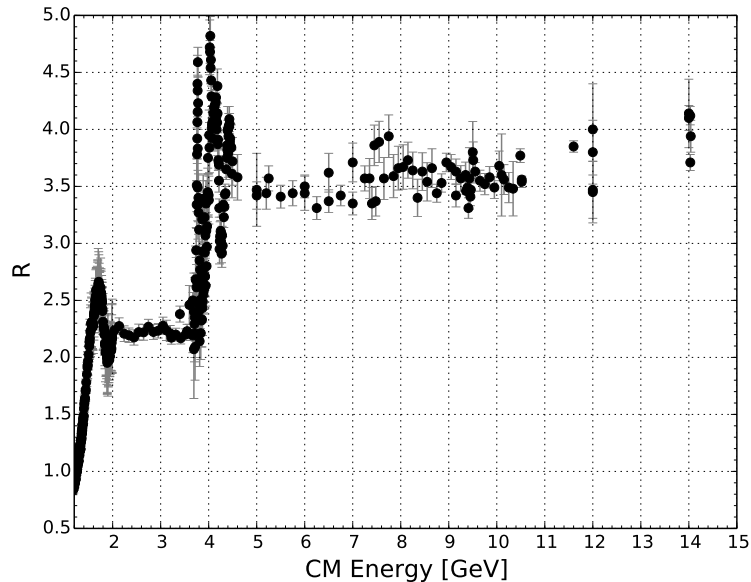
(b) (i) In an experiment the number of D^+ mesons decaying to $K^-\pi^+\pi^+$ was measured to be 7688, whereas the number decaying to $K^+\pi^-\pi^+$ was found to be 59. Draw leading-order Feynman diagrams for each of these decays. Use these measurements to calculate a value for the Cabibbo angle, explaining any assumptions that you make. [9]

(ii) No decays consistent with the process $D^+ \rightarrow e^+ + \nu$ were found. Suggest why this might have been. [2]

3. The plot below shows measured values of the ratio

$$R = \frac{\sigma(e^+ + e^- \rightarrow \text{hadrons})}{\sigma(e^+ + e^- \rightarrow \mu^+ + \mu^-)}$$

as a function of the e^+e^- centre-of-mass energy.



(a) (i) Explain the ways in which this plot provides evidence for the existence of quarks and also of colour. [8]

(ii) A resonant peak can be observed at an energy a little less than 2 GeV. Estimate the lifetime, in seconds, of the corresponding intermediate particle from the data, and comment on its value. [3]

(b) It is proposed to build an e^+e^- collider operating at centre-of-mass energies of up to 500 GeV. Describe the new processes that will modify R at energies between 15 GeV and 500 GeV. In particular you should describe quantitatively the expected change in the value of R around: (i) 90 GeV and (ii) 350 GeV, explaining any assumptions you make. Sketch a diagram of the expected behaviour of R for energies between 15 and 500 GeV. [11]

(c) Suggest why it might be practically difficult to collide electrons and positrons at these higher energies. [3]

[You may assume that the branching ratio of the Z boson to e^+e^- is 3.4%, and that the invisible branching ratio of the Z boson is 20%.]

4. (a) The non-relativistic Breit–Wigner formula for the cross-section for transitions is

$$\sigma = \pi g \left(\frac{\lambda}{2\pi} \right)^2 \frac{\Gamma_i \Gamma_f}{(E - E_0)^2 + \frac{1}{4} \Gamma^2}, \quad g = \frac{2J + 1}{(2s_1 + 1)(2s_2 + 1)},$$

where λ is the wavelength of the incident particle in the rest frame of the resonance.

(i) With reference to this equation, explain the meaning of the terms *partial width* and *total width*. State the conditions required for the validity of this equation. [4]

(ii) The cross-section for the reaction $^{11}\text{B} + \text{p} \rightarrow ^8\text{Be} + \alpha$ passes through a maximum of 160 mb at a proton laboratory energy of 1.4 MeV. The width of the resonance is 1.15 MeV and its spin and parity are given by $J^P = 1^-$. The spin of ^{11}B is $\frac{3}{2}$. Calculate two solutions for the values for the proton and α partial widths, neglecting other channels. [9]

(iii) Suggest which other decay modes of the resonance one might also expect to observe. [2]

(b) The isotope ^8Be decays with a Q value of 91 keV into two alpha particles. Explain what is meant by a *Gamow factor* G in the context of an alpha decay. Show that Q is small compared to the height of the potential barrier for alpha decay of ^8Be , and hence calculate the value of G in this decay. [10]